

Spatial Databases

Storing and querying space at scale — PostGIS, DuckDB, GeoParquet



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When file-based workflows break down

GeoPandas and file-based workflows are appropriate for in-memory analysis. They become insufficient when data exceed local memory, multiple users need shared access, or the same spatial query must run repeatedly.

A file has no query engine, no concurrency, no index that survives a restart.

A **spatial database** stores geometry, attributes, indexes, constraints, and queries together. It returns the subset needed for analysis instead of forcing each workflow to load and filter entire layers.

This module covers **the relational model, spatial indexing, and joins at scale**. GeoPandas remains the Python analysis layer; databases and analytical engines become necessary when scale, sharing, or serving change.

GeoPandas, files, and databases

GeoPandas supports in-memory spatial data science. Files support exchange and reproducibility. A spatial database adds persistent query planning, indexing, transactions, and shared access.

GeoPandas + files

- Flexible Python analysis with **GeoDataFrame**.
- Cleaning, EDA, joins, plotting, feature engineering.
- Best when data fit in memory.
- Files remain the reproducible handoff.

A spatial database

- Send **SQL**; get back only matching rows.
- Persistent **spatial index** on disk.
- Transactions + many concurrent users.
- Geometry, attributes & index in one place.

The cross-over point is **scale, sharing, and serving**. The database prepares the indexed subset; GeoPandas remains the Python layer for analysis when the result fits in memory.

What makes a database "spatial"?

A spatial database is a relational database with four spatial extensions. PostGIS, SpatiaLite, and DuckDB expose these ideas through similar SQL patterns.

1. **Spatial data types** — a **geometry** / **geography** column alongside **int** and **text**, storing points, lines and polygons.
2. **Spatial functions** — hundreds of **ST_*** functions to measure, construct, transform and relate geometries.
3. **Spatial relationships** — boolean predicates (*contains, intersects, within*) standardized by the **DE-9IM** model.
4. **Spatial indexing** — an R-tree-style index so a query touches a handful of candidate rows instead of scanning the whole table.

These four ideas explain most spatial database operations in this deck.

A one-slide SQL primer

Everything below sits on plain **relational** SQL, so here is the whole structure in one place. A **table** is typed rows × named columns; a **primary key** uniquely names each row. You *query* with one shape:

```
SELECT c.name, count(*) AS n    -- columns to return (and aggregates)
FROM   counties c              -- the table(s), with short aliases
JOIN   listings p              -- combine rows from another table...
      ON ST_Contains(c.geom, p.geom) -- ...matched by this predicate
WHERE  c.name <> 'Loving'       -- keep only rows passing a filter
GROUP BY c.name                -- collapse into one row per group
ORDER BY n DESC                -- sort the result
```

A spatial database changes only the **predicate** — **ST_Contains** where an ordinary join would write **c.id = p.county_id**. The rest is plain SQL.

The 2026 open spatial data stack

Spatial workflows usually combine several layers. GeoPandas is the Python analysis layer; files, databases, and analytical engines decide how data are stored, shared, indexed, and scanned.

Local analysis and exchange

- **GeoPandas**: in-memory vector analysis in Python.
- **GeoPackage**: robust single-file vector store.
- **GeoJSON**: web exchange, not large analytics.
- **Shapefile**: legacy interoperability only.

Query and analytics

- **PostGIS**: transactional spatial database.
- **DuckDB + GeoParquet**: local analytical SQL.
- **Sedona/Spark**: distributed spatial processing.

Columnar standards such as **GeoParquet** and **GeoArrow** connect GeoPandas, DuckDB, Spark/Sedona, and cloud storage by preserving geometry, CRS metadata, and analytical performance across tools.

Research workflow examples

Choose the data-management layer from the research task and the expected workflow, not from the name of the tool.

Research task	Practical layer	Reason
Clean census polygons and build covariates	GeoPandas + GeoPackage	Local, inspectable, reproducible vector work
Serve shared boundaries to a lab or web map	PostGIS	Concurrency, constraints, indexed queries
Scan millions of points against polygons	DuckDB + GeoParquet	Fast analytical reads without a server

The output can still return to GeoPandas when the final analysis fits in memory.

Research workflow examples (continued)

Different scale and repetition patterns call for different management layers.

Research task	Practical layer	Reason
Recompute national exposure surfaces monthly	Sedona/Spark + Parquet/Iceberg	Distributed partitions and repeated jobs
Summarize mobility at multiple spatial scales	H3-style grid + geometry checks	Fast rollups, with exact predicates where needed

These choices are not permanent. A project can move from GeoPandas to DuckDB or PostGIS when data volume, sharing, or repetition increases.

Choosing the right management layer

Choose the system from the workflow, not from tool popularity.

Small and local

- **GeoPandas** for vector exploration, cleaning, feature engineering, and small spatial joins that feed statistical or machine-learning models in Python.
- **GeoPackage** for portable course projects, field datasets, and reproducible single-file handoff.
- **GeoJSON** for lightweight web exchange, not large analytical scans.

Shared and operational

- **PostGIS** for shared editing, indexed queries, live services, constraints, and multi-user access.
- Use it when the same spatial question is asked repeatedly.

Stay in GeoPandas while the analysis is interactive, local, and model-facing.

Move from files to databases when persistence, repeated queries, shared access, or serving are required. Move to distributed systems when data or compute exceed one machine.

Choosing the analytical layer

Local analytics

- **DuckDB + GeoParquet** for fast SQL over local or remote analytical files.
- Reproducible notebooks, exploratory scans, and one-analyst projects.

Distributed analytics

- **Sedona/Spark + Parquet/Iceberg** for large partitions, repeated jobs, and distributed spatial joins.
- Use when data or compute no longer fit one machine.

Specialized indexes: H3-style grids support multiscale aggregation and rollups. Exact topology still requires geometry predicates.

What PhD students should know

Spatial data management affects reproducibility and inference, not only speed.

- **CRS and geometry metadata** must survive storage, conversion, and export.
- **Indexes and partitions** determine whether joins are feasible at scale.
- **Transactions and constraints** matter when data are edited by many people.
- **Columnar formats** improve analytical scans and cloud/object-store access.
- **Open standards** reduce dependence on a single library or vendor.
- **Provenance:** record source, processing steps, CRS, geometry repair, and simplification decisions.

Treat data layout as part of the research design.

OGC Simple Features: WKT & WKB

Before any engine, there is a **standard**. The OGC *Simple Features for SQL* specification defines the geometry types and two interchange encodings every tool agrees on — so a polygon written by PostGIS reads cleanly in DuckDB.

WKT — Well-Known *Text*, human-readable:

```
'POINT(11.34 44.49)'  
'LINESTRING(11.34 44.49, 11.35 44.50)'  
'POLYGON((0 0, 4 0, 4 4, 0 4, 0 0))'
```

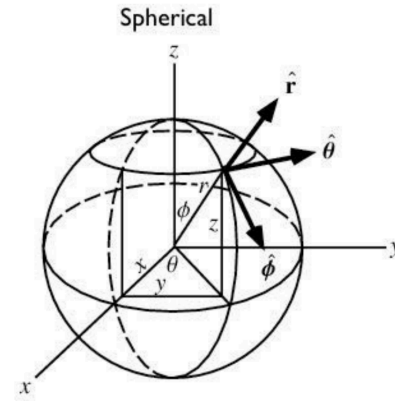
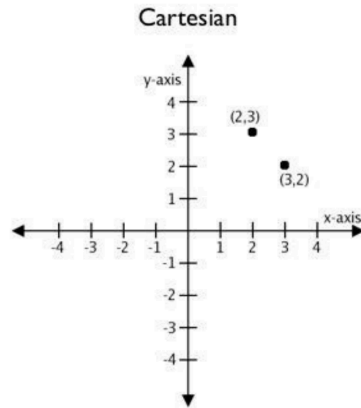
WKB — Well-Known *Binary*: the same geometry, compact and exact. It is what columns actually store.

```
-- text <=> geometry <=> binary  
SELECT ST_GeomFromText('POINT(11.34 44.49)', 4326); -- parse + SRID  
SELECT ST_AsText(geom); -- geometry back to WKT  
SELECT ST_AsBinary(geom); -- geometry to WKB
```

The number **4326** is the **SRID** — the CRS, stored *with* the data.

Geometry vs. geography

PostGIS offers **two** spatial types. **geometry** treats coordinates as points on a **flat plane** — fast, with the full function set. **geography** treats them as **lon/lat on the spheroid** and measures geodesic (shortest-path) distances on it — slower, fewer functions, but correct over long distances and across the antimeridian.



```
geometry(Point, 4326)  -- planar; fast; many ops
geography(Point, 4326) -- geodetic; metres; fewer ops
```

Rule of thumb: store as **geometry** and reproject to a metric CRS for measurement; use **geography** only for global lon/lat distances. *DuckDB is planar-only — it has no **geography** type.*

Create & load a spatial table

The SRID is not a separate column — it is a **parameter of the type**, **geometry(Point, 4326)**. Enable the extension once, declare the typed column, then build a **GiST** index so queries can use it.

```
CREATE EXTENSION IF NOT EXISTS postgis;
```

```
CREATE TABLE photos (  
  pid int PRIMARY KEY,  
  geom geometry(Point, 4326)  
);
```

```
CREATE INDEX photos_gix  
ON photos  
USING GIST (geom);
```

Getting data in:

- `shp2pgsql -s 32633 r.shp r | psql gis`
- `ogr2ogr -f PostgreSQL PG:dbname=gis x.geojson`
- `INSERT ... ST_GeomFromText(wkt, 4326)`
- `ST_GeomFromGeoJSON(...)`
- `\copy photos FROM 'p.csv' CSV HEADER`

Reproduce a server in one line: `docker run --rm -e POSTGRES_PASSWORD=pw -p 5432:5432 postgres/postgis:18-3.6`, then `CREATE EXTENSION postgis;`. (PostGIS 3.6.3, Apr 2026.)

Spatial functions: one reference slide

Hundreds of **ST_*** functions, but they fall into a handful of families. Learn the families, not the list — and remember measurement needs a **metric CRS**.

- Construct: **ST_MakePoint**, **ST_GeomFromText**, **ST_GeomFromGeoJSON**
- Inspect: **ST_SRID**, **ST_X**, **ST_GeometryType**, **ST_NPoints**
- Measure: **ST_Area**, **ST_Length**, **ST_Distance**, **ST_Perimeter**
- Validity: **ST_IsValid**, **ST_MakeValid**
- Process: **ST_Buffer**, **ST_Centroid**, **ST_Union**, **ST_Intersection**, **ST_Simplify**
- Relate: **ST_Contains**, **ST_Within**, **ST_Intersects**, **ST_DWithin**
- Reproject: **ST_Transform**, **ST_SetSRID**
- Output: **ST_AsText**, **ST_AsGeoJSON**, **ST_AsMVT**

```
-- measure in metres: reproject first, then ST_Area
SELECT name, ST_Area(ST_Transform(geom, 32633)) AS m2 FROM parks;
```

The relationships model: DE-9IM

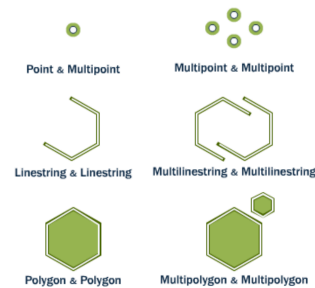
Predicates like *contains* and *touches* are not ad-hoc. Every geometry is split into **interior**, **boundary**, **exterior**; the **DE-9IM** matrix records how two geometries' parts intersect, and every named predicate is just a pattern over that 3×3 matrix. This is the OGC standard PostGIS, GEOS and DuckDB share.

- **Interior** — inside, off the boundary.
- **Boundary** — the separating edge.
- **Exterior** — everything else.

A predicate = a required intersection pattern between these parts of A and B. **ST_Relate** exposes the raw matrix.

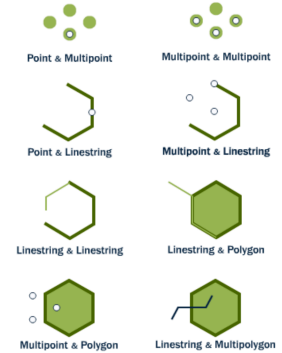
Worked example. **ST_Contains(A,B)** is the pattern **T*****FF***: A's interior must meet B's interior (**T**), B may not reach A's exterior (the two **F** s), and ***** means "don't care".

Equals



ST_Equals

Intersects

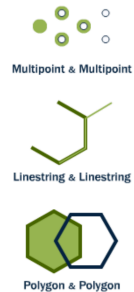


ST_Intersects

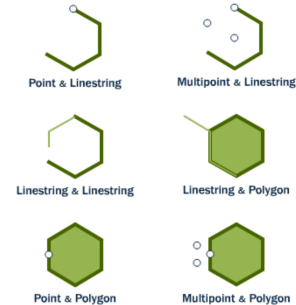
A gallery of predicates

The same matrix yields the predicates used in everyday spatial SQL. Each is one boolean **ST_*** call; the figure shows the geometric relationship behind the name.

Overlap



Touch



ST_Overlaps

<code>ST_Equals(a, b)</code>	-- same set of points
<code>ST_Intersects(a, b)</code>	-- share any point (not disjoint)
<code>ST_Contains(a, b)</code>	-- b lies fully inside a
<code>ST_Within(a, b)</code>	-- a lies fully inside b

ST_Touches

<code>ST_Touches(a, b)</code>	-- meet only on boundaries
<code>ST_Overlaps(a, b)</code>	-- partial, same dimension
<code>ST_Crosses(a, b)</code>	-- partial, lower dimension
<code>ST_DWithin(a, b, d)</code>	-- within distance d

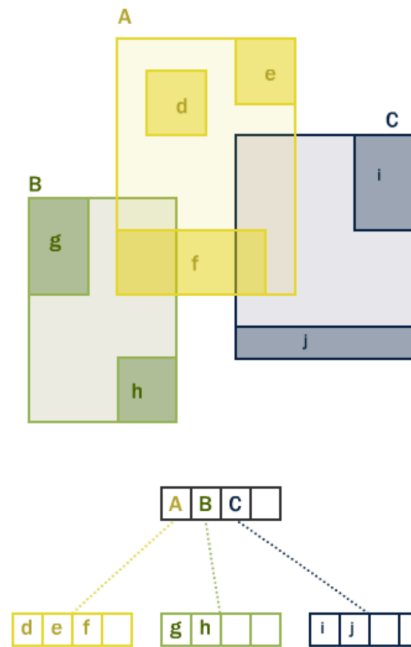
Spatial indexing: GiST & the R-tree

This is the heart of the module. Without an index, a predicate compares every row against every other — $O(N \times M)$. The fix is an **R-tree**: a hierarchy of nested **bounding boxes** that lets a query discard whole branches at once.

Boxes nest into parents (A, B, C); leaves hold the real geometries (d...j). A search descends only the boxes its query box touches.

In PostGIS the implementation is **GiST** (**USING GIST**); the R-tree is the *concept*, GiST the *mechanism*. DuckDB writes it as **USING RTREE**.

Same idea as Deck 02's GeoDataFrame **.sindex** — here it is pushed into the database and persisted.



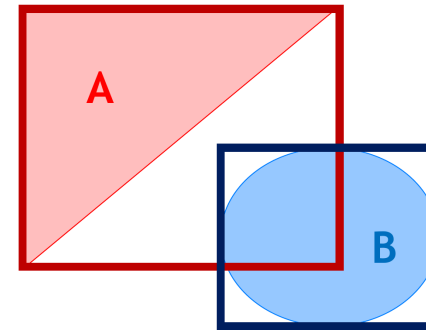
Two-pass: filter, then refine

Bounding boxes are approximate filters: boxes can overlap while the true shapes do not. Spatial queries therefore run in **two passes** — the index (**&&**) cheaply returns *candidates* whose boxes overlap, then the exact topological test (**ST_Intersects**) confirms the real answer on that short list.

Here A's box and B's box overlap, so **A && B = TRUE** — yet the triangle and circle never meet, so the exact test is **FALSE**.

Only **bounding-box operators** (**&&**) reach the index directly. Functions such as **ST_Intersects** and **ST_DWithin** include that filter internally, so write the exact predicate and let the planner use the index.

The figure labels the exact test **_ST_Intersects** — PostGIS's internal predicate that runs after the **&&** filter.



A && B = TRUE

_ST_Intersects(A && B) = FALSE

* _ST_Intersects: geometric intersection

Writing index-aware queries

The planner needs a bounding-box operator to use the GiST index. The reliable pattern is **ST_DWithin** (not **ST_Distance(...) < d**) with the indexed column left unchanged on its side of the predicate.

```
-- "within 500 m of a point": ST_DWithin uses the index;  
-- ST_Distance(geom, p) < 500 would NOT.  
SELECT name  
FROM    roads                                -- geom stored in EPSG:32633 (metres)  
WHERE    ST_DWithin(  
        geom,  
        ST_Transform(  
            ST_SetSRID(ST_MakePoint(11.34, 44.49), 4326), 32633),  
        500);                                -- 500 metres, because geom is metric
```

Wrap the *constant* in **ST_Transform**, never the indexed column — transforming **geom** would defeat the index on every row.

Spatial joins in SQL

A **spatial join** matches rows from two tables by a geometric predicate instead of a shared key. The index is what makes this operation scale from a small example to millions of candidate geometries.

```
-- Point-in-polygon: how many listings fall in each county?  
SELECT c.name, count(*) AS n  
FROM    counties c  
JOIN    listings p ON ST_Contains(c.geom, p.geom)  
GROUP BY c.name  
ORDER BY n DESC;
```

```
-- Proximity join: every hospital within 2 km of each school  
SELECT s.id, h.id  
FROM    schools s  
JOIN    hospitals h ON ST_DWithin(s.geom, h.geom, 2000); -- metres
```

Lesson — a spatial join is a join keyed on a *geometric predicate* —
ST_Contains for *inside*, **ST_DWithin** for *near* — run in the database, at scale.

PostGIS from Python — the bridge

You rarely choose *either* PostGIS or GeoPandas — you move between them. GeoPandas speaks PostGIS directly, so the database does the heavy spatial filter and pandas does the rest.

```
import geopandas as gpd
from sqlalchemy import create_engine

engine = create_engine("postgresql://user:pass@localhost/gis")

# DB -> GeoDataFrame (let PostGIS do the spatial work first)
gdf = gpd.read_postgis(
    "SELECT name, geom FROM counties", engine, geom_col="geom")

# GeoDataFrame -> DB
gdf.to_postgis("counties", engine, if_exists="replace")
```

The same bridge to the analytical world is a **GeoParquet** file: GeoPandas and DuckDB both read and write it, so it is their lingua franca — next.

Pivot: from serving to analytics

PostGIS is a **transactional server** — superb at serving a live map and guarding writes from many users. But "scan 200 million points and aggregate" is a different job: **analytical (OLAP)**, columnar, single-analyst, often read-only.

DuckDB provides an in-process, columnar, vectorized engine with a **spatial** extension. It uses familiar SQL and **ST_*** predicates, but is designed for large analytical scans rather than multi-user transactional serving.

- In current DuckDB, **GEOMETRY** is available in the core engine; the **spatial** extension adds the **ST_*** functions, **ST_Read**, and GeoParquet I/O.
- No server, no **CREATE EXTENSION** ceremony — **pip install duckdb** and go.
- It can **query files in place**, so the load step often disappears.

DuckDB hands-on (1): connect & build tables

A real session — `pip install duckdb`, no server. The `GEOMETRY` type is core; we `LOAD spatial` only for the `ST_*` functions and `ST_Read`. We rebuild exactly Deck 02's Texas example: counties (polygons) × Airbnb listings (points).

```
import duckdb
con = duckdb.connect()                # in-process, no server
con.execute("INSTALL spatial; LOAD spatial;") # ST_* + ST_Read

-- polygons straight from a GeoJSON file via ST_Read
CREATE TABLE counties AS
  SELECT NAME AS county, geom FROM ST_Read('data/texas/texas.geojson');

-- points built from lon/lat columns of a CSV (run from notebooks/)
CREATE TABLE listings AS
  SELECT id, ST_Point(longitude, latitude) AS geom
  FROM read_csv_auto('data/texas/listings.csv.gz');
-- counties: 254    listings: 5835
```


DuckDB hands-on (2): the spatial join

The same point-in-polygon join as PostGIS, on the same data — it returns **exactly Deck 02's GeoPandas result** (Travis = 5792). One engine, one query, zero server.

```
SELECT c.county, count(*) AS n
FROM   counties c
JOIN   listings p ON ST_Contains(c.geom, p.geom)
GROUP BY c.county
ORDER BY n DESC
LIMIT 3;
```

county	n
Travis	5792
Williamson	35
Hays	7

-- 5834 of 5835 points matched a county (one stray geocode falls outside).

Lesson — the same analysis lives in SQL: DuckDB reproduces deck 02's join with no server and no Python loop — files in, answer out

DuckDB hands-on (3): index & GeoParquet

Two more real outputs. DuckDB's index is **USING RTREE** (same concept, PostGIS calls it GiST). And a **GeoParquet** round-trip: the columnar file format that is the interchange currency between DuckDB and GeoPandas.

```
CREATE INDEX idx ON listings USING RTREE (geom); -- persists with the DB
-- the R-tree fires for ST_Within/ST_Intersects/ST_Contains over a
-- constant – NOT for ST_DWithin (unlike PostGIS GiST). 0.05 deg ~ 5.5 km.
SELECT count(*) FROM listings
WHERE ST_Within(geom,
  ST_Buffer(ST_Point(-97.7431, 30.2672), 0.05));
-- -> 3920 listings near central Austin (lon/lat; DuckDB is planar)
```

```
-- write GeoParquet, then read it straight back
COPY (SELECT id, geom FROM listings) TO 'listings.parquet';
SELECT count(*), ST_GeometryType(geom) FROM 'listings.parquet'
GROUP BY 2; -- -> 5835 | POINT
-- file carries a 'geo' metadata key -> GeoPandas reads it as a GeoDataFrame
```

Query files in place

With **https**, DuckDB can query **remote GeoParquet without downloading the entire file**. Only the bytes required by the query need to move, so the load step becomes optional for many analytical workflows.

```
INSTALL https; LOAD https;           -- read over HTTPS / S3

-- count features straight out of a remote GeoParquet (executed):
SELECT count(*), ST_GeometryType(geometry) AS g
FROM
'https://github.com/opengeospatial/geoparquet/raw/main/examples/example.parquet'
GROUP BY 2;
-- -> 3 MULTIPOLYGON, 2 POLYGON  (read in place, nothing saved locally)
```

```
# real Overture pattern (S3); keep a local file as offline fallback
src = "s3://overturemaps-us-west-2/release/2026-.../*.parquet"
# con.execute(f"SELECT ... FROM read_parquet('{src}') WHERE bbox...")
```

GeoParquet is **mid-transition**: 1.1 (WKB + metadata, what tools emit today)
→ native Parquet geometry → GeoParquet 2.0.

Distributed spatial analytics

When data no longer fit a single machine or analysis must run repeatedly over large partitions, the workflow moves from local SQL to distributed spatial SQL.

Typical stack

- Object storage with Parquet / GeoParquet
- Table format such as Iceberg or Delta
- Spark with Apache Sedona for spatial SQL
- Partitioning by space, time, or both

Use when

- Data volume is beyond local memory and disk
- Many jobs scan the same archive
- Spatial joins need distributed execution
- Outputs feed reproducible research pipelines

The tradeoff is operational complexity: distributed systems require stronger discipline around partitioning, metadata, schemas, and reproducibility.

Spatial indexes beyond R-trees

R-trees accelerate geometry predicates. Global grid indexes solve a different problem: they assign locations to stable cells that can be grouped, joined, and aggregated efficiently.

- **H3**: hierarchical hexagonal grid; common for global aggregation, mobility, exposure summaries, and multiscale rollups.
- **S2 / geohash**: alternative global cell systems with different geometry and indexing properties.
- **Use carefully**: cell indexes approximate geometry. Resolution choice affects bias, boundary effects, and privacy.

Grid indexes are data-management tools. They do not replace geometry predicates when exact topology matters.

PostGIS vs. DuckDB: when to use which

They solve different jobs. Use the transactional server to *serve*; use the analytical engine for large read-only analysis. GeoParquet is the bridge between them.

Use PostGIS when...

- Many users read *and write* concurrently.
- You serve a live app / web map.
- You need transactions & constraints.
- You need geodetic **geography**.

Use DuckDB when...

- One analyst, big read-only scans.
- Data already lives in Parquet / on S3.
- You want zero-setup, in-process SQL.
- You want to query files in place.

Same SQL, same **ST_*, same R-tree.** Learn the predicates once; switch engines as the job demands — or use **ibis** to write one query for either.

Four-level decision framework

Level	Use when	Avoid when
File	Small, portable, single-user work	Repeated queries or concurrent edits
PostGIS	Shared, indexed, transactional data	One-off local scans dominate
DuckDB + GeoParquet	Fast analytical SQL on local/cloud files	Multi-user writes are required
Sedona/Spark	Distributed joins and repeated large scans	Data fit comfortably on one machine

For research, start with the simplest layer that preserves correctness and reproducibility. Move up only when scale, sharing, or serving requires it.

Summary

- A **spatial database** = SQL + four extensions: **types**, **functions**, **predicates (DE-9IM)**, and **indexing**.
- **PostGIS** is the transactional reference; **geometry** (planar) vs **geography** (geodetic); SRID is a **type parameter**, not a column.
- **GiST / R-tree** + the **two-pass** filter-and-refine make spatial **joins** scale — the whole point of a spatial DB.
- **DuckDB + GeoParquet** is the analytical tier: in-process, columnar, and able to **query files in place**.
- **GeoPandas** remains the **Python modeling layer**: use databases and engines to prepare the right subset, then return to GeoDataFrames for analysis.
- **Sedona/Spark + Parquet/Iceberg** is the distributed tier for repeated large-scale spatial analytics.
- **H3-style indexes** support multiscale aggregation, but exact topology still requires geometry predicates.

Spatial data management is a modeling foundation: the storage layer determines which joins, measurements, validations, and reproducibility checks are feasible.

Further learning

- **PostGIS documentation:** <https://postgis.net/documentation/>
- **Intro to PostGIS workshop:** <https://postgis.net/workshops/postgis-intro/>
- **DuckDB spatial:** https://duckdb.org/docs/stable/core_extensions/spatial/overview
- **GeoParquet:** <https://geoparquet.org/>
- **Apache Sedona:** <https://sedona.apache.org/>
- **H3:** <https://h3geo.org/>
- **GeoArrow:** <https://geoarrow.org/>